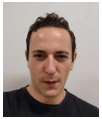
AI-Powered Vibe Coding with GitHub Copilot

AI Workflows for Biomedical Data Science



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HARVARD MEDICAL SCHOOL | BLAVATNIK INSTITUTE BIOMEDICAL INFORMATICS

Workshop Materials

Everything we do today is open-source.

- Slides
- Setup Guides
- Live Demo Scripts
- Hands-On Datasets & Starter Code

GitHub Repository:

`https://github.com/ccb-hms/ai-powered-coding`

Workshop Agenda (120 Minutes)

Part 1: Presentation & Core Concepts **30 min**

The AI Landscape • Context Memory • Prompting Strategies • Data Privacy

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Part 2: Live Demonstration **45 min**

Copilot in Action • Code Refactoring • Error Debugging • IDE Tools

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Part 3: Hands-On Exercise **45 min**

TCGA Survival Analysis • Guided AI Challenges • Instructor Support

What we *mean* by "Vibe Coding"

"Focusing on the **overall outcome** *(the vibe) rather than writing syntax line-by-line."*

— Coined by Andrej Karpathy (2025)

The Approach:

- **Natural Language:** Prompting AI to generate, debug, and iterate.
- **Rapid Prototyping:** Prioritizing speed and creativity to test ideas quickly.
- **Role Shift:** The developer becomes a **director**, not a typist.

What we *don't* mean by "Vibe Coding"

The Golden Rule of AI Coding

You cannot just "go with the vibe." You remain **strictly accountable** for understanding and validating all AI-generated code.

Important Limitations:

- **No Blind Acceptance:** Never run code you cannot read or explain.
- **Standards Still Apply:** Security and best practices remain mandatory for publication.
- **Prototyping $\neq$ Production:** Great for fast ideas, but real software still demands engineering rigor.

The Vibe Coding Mindset: Offloading vs. Surrender

The Goal: Focus on scientific intent, not syntax.

Cognitive Offloading (GOOD)

- **AI is the typist.**
- Offload boilerplate, syntax, and formatting.
- Frees mental space for high-level scientific logic.

Cognitive Surrender (BAD)

- **AI is the scientist.**
- Blindly trusting AI-generated math or logic.
- Running code you cannot read or explain.

You remain the pilot. Copilot is just the navigator.

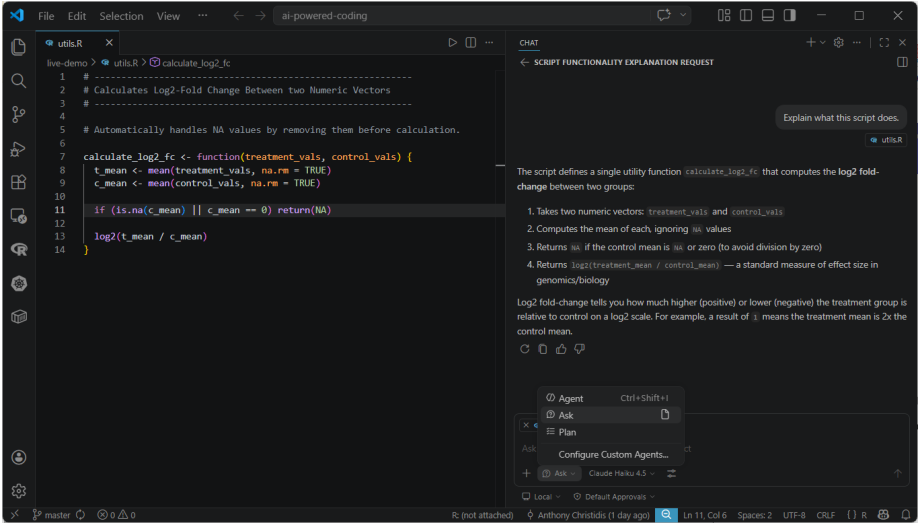
What is GitHub Copilot Today?

It is no longer just "autocomplete." It is an umbrella of integrated tools:

1. **Code Completion** (Ghost Text)
2. **Copilot Chat** (Sidebar & Inline)
3. **Copilot Agents** (Autonomous edits)

Plus: You can now choose your underlying AI model (Claude Sonnet 4.6, GPT-5.4, etc.).

The Copilot Interface



The AI Coding Landscape

Copilot is the enterprise standard, but it is not alone.

Cursor

A standalone, AI-first editor. Currently the hottest competitor for power users.

GitHub Copilot

The enterprise & university standard extension. **New:** Now supports running local, offline models for strict HIPAA privacy.

Continue (.dev)

Open-source, highly customizable extension. Easily connects to any custom cloud API or local model (Ollama).

The Old Way: Browser Chat (ChatGPT / Claude Web)

- **Isolated:** The AI has no idea what is on your computer.
- **Tedious:** Requires constant copy/pasting of your code, column names, and errors.
- **Inaccurate:** Frequently hallucinates variable names because it lacks context.

The Old Way: Manual Context

Good afternoon, Anthony Christidis

Hello! I'm GPT-5.1, running in the Harvard AI Sandbox. I have a 400k token context window (~300k words) and my knowledge cutoff is September 2024. I also have the ability to search the web in real time to access the most up-to-date information, something that most other models may not support. How can I help you today?

I need to analyze my biomarker data and plot the results.

Here are the columns in my biomarker_dummy.csv file:
patient_id, age, treatment_group, biomarker_score, response_status.

Here is my utils.R script:

```
# [Pretend you copied and pasted all 30 lines of your utils.R code here]
# (Imagine a long block of code here calculating summary stats)
```

And here is my current script, live_demo.R:

```
library(tidyverse)
source("utils.R")
df <- read_csv("biomarker_dummy.csv")
```

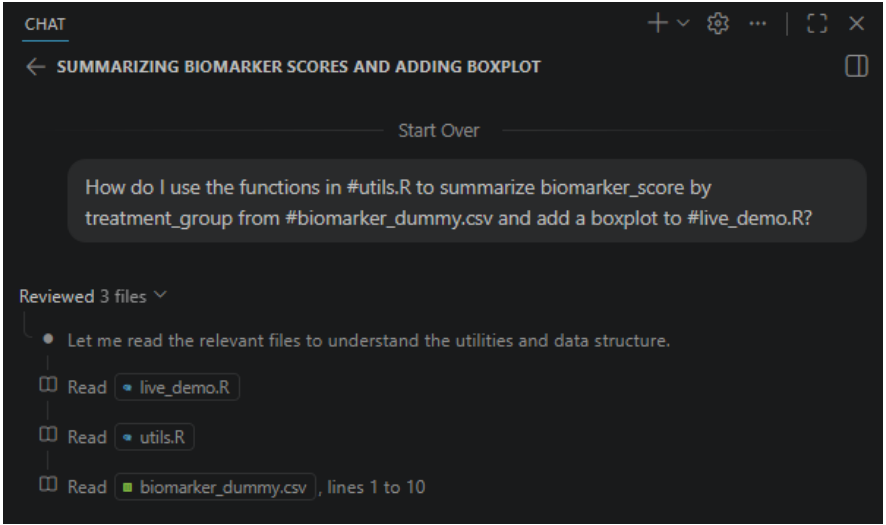
How do I use the helper functions from my utils script to calculate the mean biomarker score for each treatment group, and then make a boxplot of it in my live_demo script?

File Search

The New Way: IDE Integration (GitHub Copilot)

- **Context-Aware:** The AI lives *inside* your workspace.
- **Omniscient:** It automatically sees your open tabs, folders, and highlighted text.
- **Accurate:** Writes code that perfectly matches *your* environment and variable names.

The New Way: Automatic Context



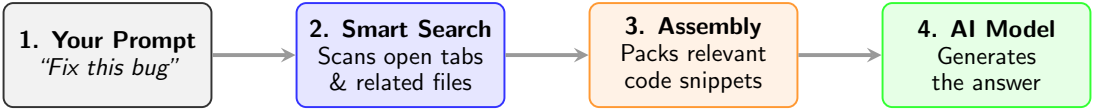
How Copilot Works: The Context Window

- **Myth:** *"Copilot reads my entire project every time I type."*
- **Reality:** While modern AI has huge memory, feeding it thousands of irrelevant files causes hallucinations and slows it down.

The Solution: Copilot selectively curates a limited **"Context Window"** to stay fast and accurate.

How Copilot Works: Smart Search

Copilot acts as an incredibly intelligent search engine:



The AI is remarkably good at finding the right files automatically.
But when it occasionally misses one, we take manual control!

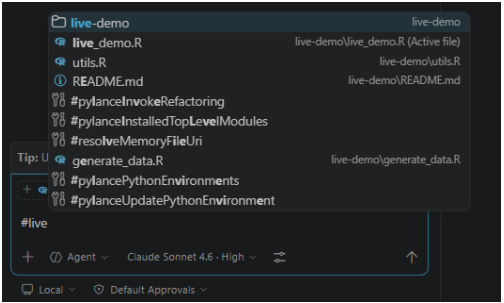
Best Practice: Pinning Context with

When automatic search needs a hint, take manual control.

Use the # symbol to explicitly "pin" exact files into the AI's memory.

The Most Powerful Tags:

- `#filename` (e.g., `#utils.R`)
- `#selection` (Only reads highlighted text)
- `@terminal` (Reads your error messages!)



Typing # instantly opens the context menu

Interacting with AI: Three Levels of Control

- **1. Chat (Read-Only):** The AI acts as a tutor. It explains concepts and generates code snippets for you to manually copy and paste.
- **2. Inline Edit (`Ctrl+I`):** The AI acts as a typist. It rewrites your highlighted code in place and provides a Red/Green diff for you to accept.
- **3. Agent Mode:** The AI acts as a junior developer. It autonomously creates new files, edits multiple scripts at once, and can run terminal commands.

Choosing Your Environment

VS Code

- The industry standard.
- Offers the most stable, full-featured Copilot experience.

Positron

- Uses *Positron Assistant*.
- **Superpower:** Reads your live R/Python environment (dataframes!), not just text.

RStudio

- **Stable Release:** Copilot autocomplete only.
- **Daily Builds:** The full Posit AI Assistant was just added in early beta!

Upcoming Workshop Announcement

Posit AI Tools @ Harvard

Advanced AI Integration in R and Python

- **Date:** April 29th
- **Time:** 2:00 PM – 4:00 PM
- **Location:** Lahey Room

Phil Bowsher

Director, Health & Life Sciences
Industry Leader

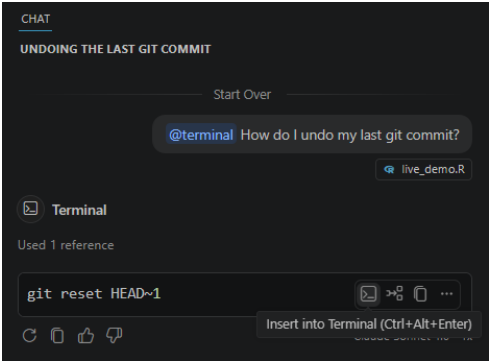
Posit (formerly RStudio)

Terminal Productivity

Copilot manages your environment, not just your script.

Terminal Assistance

- Use `@terminal` to ask bash, shell, or git questions directly in chat.
- Copilot provides a "Run in Terminal" button to execute the command instantly.

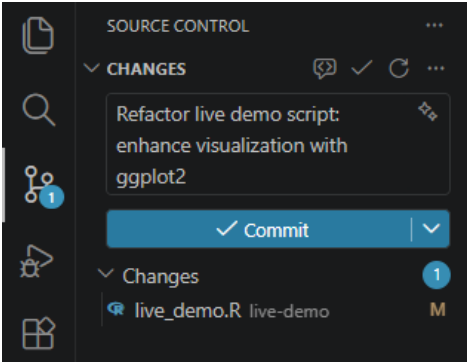


Git Productivity: The “Magic Commit”

Stop writing “fixed bug” in your commit history.

Automated Commits

- Clicking the **sparkle** icon allows Copilot to read your exact file diffs.
- It auto-generates a highly professional, accurate commit message summarizing your work.



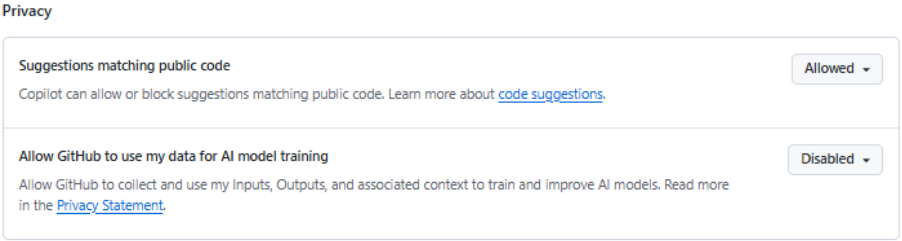
Data Privacy & Security at HMS

Critical Privacy Rule

NEVER use GitHub Copilot (or any cloud-based LLM) with Protected Health Information (PHI), Personally Identifiable Information (PII), or restricted clinical datasets.

Action Required:

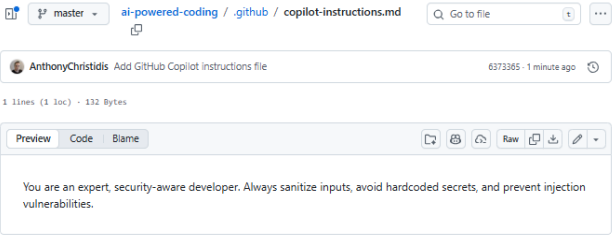
- Go to GitHub Settings → Copilot.
- **DISABLE:** *"Allow GitHub to use my data for AI model training."*



Safer Vibe Coding: The Power of "Rules Files"

AI generates insecure code by default. We must explicitly instruct it to be safe.

- **The Risk:** Studies show up to 70% of raw AI-generated code contains vulnerabilities.
- **The Fix: Rules Files** (e.g., `copilot-instructions.md`).
- **How it works:** These files silently attach global instructions (*"You are a security-aware developer"*) to every prompt, dropping weaknesses by over 40%.



Resource:

Wiz: Open-Source Security Rules Files

Beyond Copilot: AI *Inside* Your Code

AI can also act as a computational engine within your scripts.

LLMs as R/Python Functions:

- **API Endpoints:** Call models (Claude, GPT) directly from your script.
- **AI Data Manipulation:** Packages like `ellmer` (Posit) tidy data or run NLP on dataframes using plain English.

Resource:

Comprehensive List of R LLM Packages

Want to learn this?

Posit AI Tools @ Harvard: Advanced AI Integration in R and Python

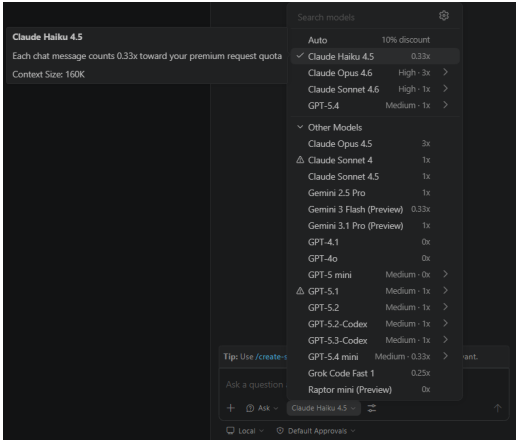
April 29th — Lahey Room

Preparing for the Hands-On: Model Selection

You are not locked into one AI. You can swap engines on the fly.

Action Required:

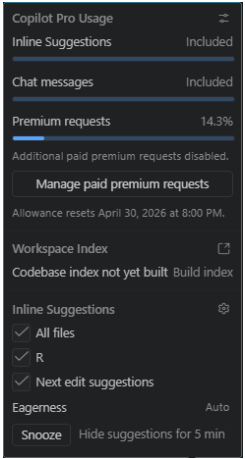
- Open Copilot Chat.
- Click the **Model Dropdown** in the top corner.
- Select your preferred model.



Understanding Your Monthly Quota

The Multiplier System:

- **Base Models:** Unlimited usage.
- **1x Models:** Standard burn rate.
- **3x Models:** Burns through your monthly limit 3 times faster (Expensive!).
- **0.33x Models:** Highly efficient. Gives you 3x *more* messages per month.



Keep an eye on your usage dashboard!

Our Experience: Picking the Right Model

Claude Haiku 4.5 *0.33x Cost*

Our Take (The Daily Driver): Fast and handles 80% of everyday coding and debugging. Excellent for stretching your monthly quota. **Our recommendation for today's hands-on!**

Sonnet 4.6 & Gemini Pro 3.1 *1x Cost*

Our Take (The Sweet Spot): The best overall balance of intelligence and cost for data science.

Claude Opus 4.5 *3x+ Cost*

Our Take (The Heavyweight): Extremely smart, but burns through your monthly limit very quickly. Save it for massive, complex architecture problems.

Live Demonstration

Let's see it in action.